

Data Structure (C Programming)

25/08/2026

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Roll No : 15

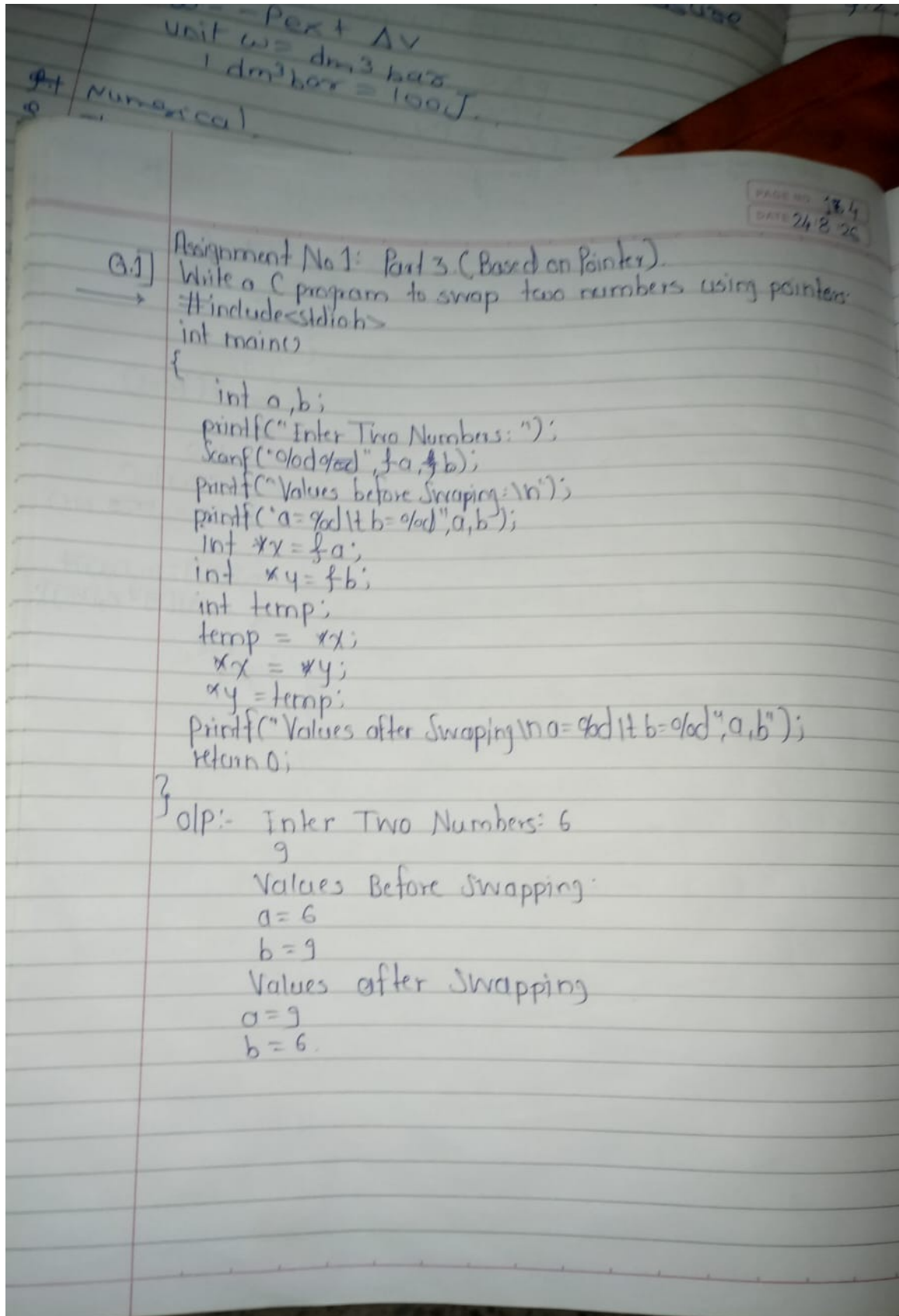
Batch : B1

Prerequisite

Based on Pointers.

Q1. Write a C program to swap two numbers using pointers.

Input:



Output:

```
student@student-OptiPlex-SFF-7020:~/Pictures/Harshad$ gcc -o output pointer1.c
student@student-OptiPlex-SFF-7020:~/Pictures/Harshad$ ./output
Enter Two Numbers:6
9
Values Before Swapping:
a=6
b=9
Values After Swapping :
a=9
b=6
student@student-OptiPlex-SFF-7020:~/Pictures/Harshad$
```

Q2. Write a C program to find the length of a string using pointers (without using `strlen`).

Input:

Q.2] Write a C Program to find the length of a String using Pointers (without using `strlen`).

```
#include <stdio.h>
void main()
{
    char Str[50];
    int i=0, len=0;
    printf("\n Enter the String: ");
    gets(Str);
    while(Str[i] != '\0')
    {
        len++;
        i++;
    }
    printf("The length of string is %d", len);
}
```

Output:

```
student@student-OptiPlex-SFF-7020:~/Pictures/Harshad$ gcc -o output pointer1.c
student@student-OptiPlex-SFF-7020:~/Pictures/Harshad$ ./output
Enter the String: Harshad

The Length of the Harshad
is 7
student@student-OptiPlex-SFF-7020:~/Pictures/Harshad$
```

Q3. Write a C program to access and display array elements using pointers.

Input:

Q3] Write a C Program to access and display array elements using pointers.

```
→ #include <stdio.h>
int main()
{
    int arr[5];
    int *p = arr;
    printf("Enter Array elements:\n");
    for(int i=0; i<5; i++)
    {
        scanf("%d", (p+i) arr[i]);
    }
    for(int i=0; i<5; i++)
    {
        printf("%d", *p(p+i));
    }
    return 0;
}
```

Name: Harshad Teli
Roll No: 15
Sign: Harshad

Output:

```
student@student-OptiPlex-SFF-7020:~/Pictures/Harshad$ ./output
Enter Two Numbers:12
6

The GCD is 6
student@student-OptiPlex-SFF-7020:~/Pictures/Harshad$
```

